



University of Khartoum

Faculty of Mathematical Sciences and Informatics

GridGuard Khartoum

AI-Based Smart Load Shedding and Energy Management System for Developing Countries

A graduation project presented to the Faculty of Mathematical Sciences and Informatics, University of Khartoum, in partial fulfilment of the requirements for the degree of Bachelor of Computer Science.

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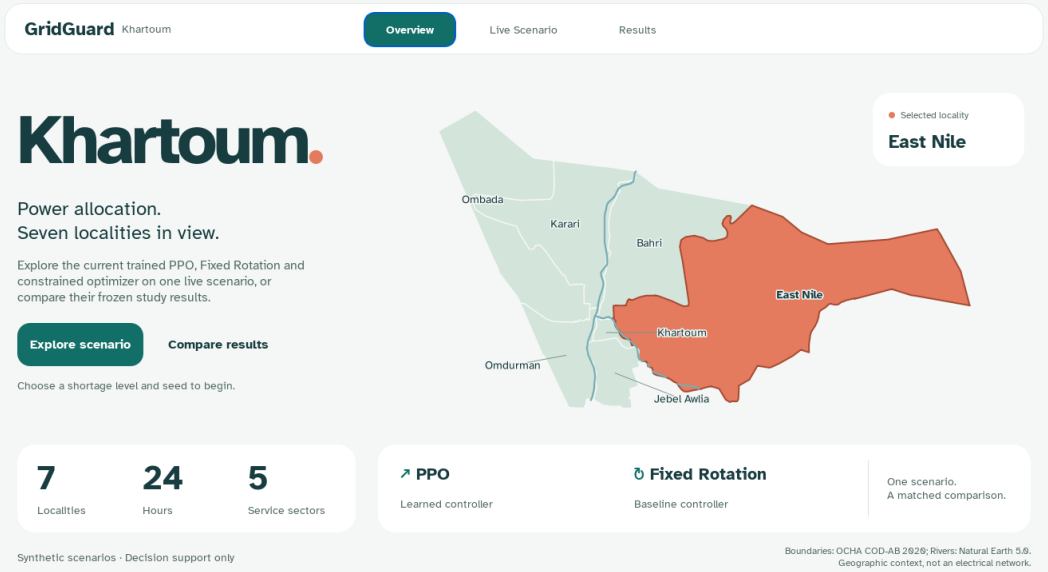
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October 2026

Hourly shortage allocation in Khartoum

Seven localities, five sectors and a shared supply pool

- Protect Hospital, Water and Bakery demand.
- Allocate remaining supply to Residential and Commercial demand.
- Measure service, interruption duration and distribution efficiency.



All demand and supply traces are synthetic. The map shows administrative geography.

Context: World Bank (2025). Related control research: Chen et al. (2023).

The simulation and service choices

24 hourly decisions with current observations and accumulated service history

Service tier	Requested demand
Off	No sectors
Critical Only	Hospital, Water and Bakery
Full Service	All five sectors

Shortage	Supply construction
Mild	80% of total demand
Moderate	65%, minus 10 pp at hours 17–20
Severe	50%, minus 15 pp at hours 16–21

- Independent demand and supply perturbations within ±5%.
- One supply pool. No network power flow, storage or restoration transients.

Critical protection through feasible actions

PPO and Constrained Optimization share 128 Critical Only / Full Service combinations

Critical reservation

$$C + \sum_{i=1}^{7} x_i b_i \leq S, \qquad x_i \in \{0, 1\}$$

C: total critical demand b_i: locality noncritical demand S: available supply

$$\alpha = \min\left(1, \frac{S}{\max(Q, \varepsilon)}\right), \qquad \text{delivered} = \alpha \, \text{requested}$$

Shared proportional dispatch uses $\varepsilon = 10^{-8}$. If critical demand cannot fit, the all-Critical-Only fallback still records the resulting interruption.

Masking: Huang and Ontañón (2022). Safety distinction: García and Fernández (2015).

Three original decision procedures

The optimizer and PPO use the same current information and feasible set

Controller	Decision rule	Horizon
Fixed Rotation	Three localities Full Service for each three-hour block	Fixed schedule
Constrained Optimization	Checks 128 assignments and maximizes current reward over the feasible set	Current hour
PPO	Chooses the highest-logit feasible action from a learned policy	Daily training objective

$$a_t^{\text{opt}} = \arg\max_{a \in \mathcal{A}(s_t)} r_t(s_t, a), \qquad \pi^{\text{PPO}} : \ \max \mathbb{E}\left[\sum_{t=0}^{23} r_t\right]$$

Reward and public service measures

The training objective combines protection, utilization, fairness and switching

$$r_t = \frac{5C_t + 2U_t + 2J_t - 0.25W_t - 3O_t}{9}$$

C: hourly critical service U: supply utilization J: cumulative-service fairness
W: fraction changing tier O: excess requested power / total demand

$$J_t = \frac{\left(\sum_{i=1}^{7} z_{i,t}\right)^2}{7\sum_{i=1}^{7} z_{i,t}^2 + \varepsilon}$$

z_i,t is cumulative served energy / cumulative demand. Report mean hourly Jain fairness, direct duration measures and energy separately from daily reward.

Fairness definition: Jain, Chiu and Hawe (1984).

Definitions used in the hourly reward

All reward components are dimensionless. Power is measured in MW and energy in MWh.

$$C_t = \frac{\sum_i \sum_{k \in \mathcal{K}_c} y_{i,k,t}}{\max\left(\sum_i \sum_{k \in \mathcal{K}_c} d_{i,k,t}, \varepsilon\right)}, \qquad U_t = \frac{Y_t}{\max(\min(S_t, D_t), \varepsilon)}$$

D: total demand Q: requested power Y: delivered power S: supply

$$z_{i,t} = \frac{E^-_{i,t+1}}{\max(D^-_{i,t+1}, \varepsilon)}, \qquad J_t = \frac{\left(\sum_i z_{i,t}\right)^2}{7 \sum_i z_{i,t}^2 + \varepsilon}$$

$$W_0 = 0, \quad W_t = \frac{1}{7} \sum_i \mathbf{1}[a_{i,t} \neq a_{i,t-1}] \ (t > 0), \quad O_t = \frac{\max(0, Q_t - S_t)}{\max(D_t, \varepsilon)}$$

Fairness uses cumulative service after dispatch. Switching is zero at hour 0. ε = 10^{-8} protects denominators; it adds no physical supply.

State, observations and transitions

A 24-hour finite-horizon decision process with action-independent demand and supply

$$X_t = (t, d_t, S_t, D_t^-, E_t^-, a_{t-1}, \xi_t), \qquad t = 0, \ldots, 23$$

ξ is the scenario-generator state. The actor observes a 46-feature projection, without access to the future trace.

D⁻ / E⁻: demanded / served MWh before the hour. a: locality service tier.

$$D_{i,t+1}^- = D_{i,t}^- + \Delta t \sum_k d_{i,k,t}, \qquad E_{i,t+1}^- = E_{i,t}^- + \Delta t \sum_k y_{i,k,t}$$

Δt = 1 hour. The action updates served-energy history and the previous tier. The scenario supplies the next demand and availability.

Cumulative energy starts at zero. Episodes terminate after hour 23 with zero terminal value.

Masked PPO training

Separate actor and critic networks: 46 inputs, two 256-unit tanh layers, 128 actor logits

$$\rho_t(\theta) = \frac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)}$$

$$\mathcal{L} = -\mathbb{E}_t\left[\min\left(\rho_t \widehat{A}_t, \text{clip}(\rho_t, 0.8, 1.2)\widehat{A}_t\right)\right] + 0.25\mathbb{E}_t[(V_\phi - \widehat{R}_t)^2] - c_H\mathbb{E}_t[H(\pi_\theta)]$$

The same stored feasibility mask defines both policy probabilities.
Advantages normalize over the full rollout. Value targets remain fixed.

γ = 1.0, GAE λ = 0.95, zero terminal value. The critic term is exactly 0.25 MSE. Entropy coefficient decreases from 0.01 to 0.001.

PPO: Schulman et al. (2017). Masking: Huang and Ontañón (2022).

Five independent runs and frozen selection

Fresh initialization, disjoint training streams and a fixed training budget

Training element	Configuration
Training seeds	1729, 2718, 3141, 5772, 8111
Each run	3,000 iterations × 256 episodes × 24 hours
Total PPO exposure	92,160,000 transitions
Initialization	Fairness Rule demonstrations: 32,760 states, 10 epochs
PPO updates	4 epochs, minibatch 1,024, learning rate 0.0003 to zero

Select the best validation reward among checkpoints with zero critical/power violations and fairness/utilization at least Fixed Rotation overall and at every shortage level.

Matched evaluation and uncertainty

All final-test decisions follow the frozen five-run validation selection

Split	Independent base draws	Coupled conditions
Validation	5 seeds × 32 = 160	480
Final test	10 seeds × 32 = 320	960

- 12 methods: Fixed Rotation, optimizer, five selected PPO actors and five iteration-600 actors.
- 11,520 trajectories. Bootstrap resamples five training runs and matched base-draw clusters.
- 10,000 resamples. Two primary claims use adjusted 97.5% intervals.

Statistical reliability: Agarwal et al. (2021).

Original criterion against Fixed Rotation

Every requirement must hold for the declared system-level improvement claim.

Requirement	Exact rule
Adjusted overall intervals	Critical-interruption upper bound < 0; fairness and utilization lower bounds > 0
Every shortage level	Mean interruption difference ≤ 0; mean fairness and utilization differences ≥ 0
Every selected PPO trajectory	Zero critical interruptions, outages and power-balance violations
Validation eligibility	All five selected policies eligible under the frozen selection rule

Effects are PPO minus Fixed Rotation. Equal-run and equal-condition means; adjusted 97.5% intervals for the two primary comparisons.

Original criterion against Constrained Optimization

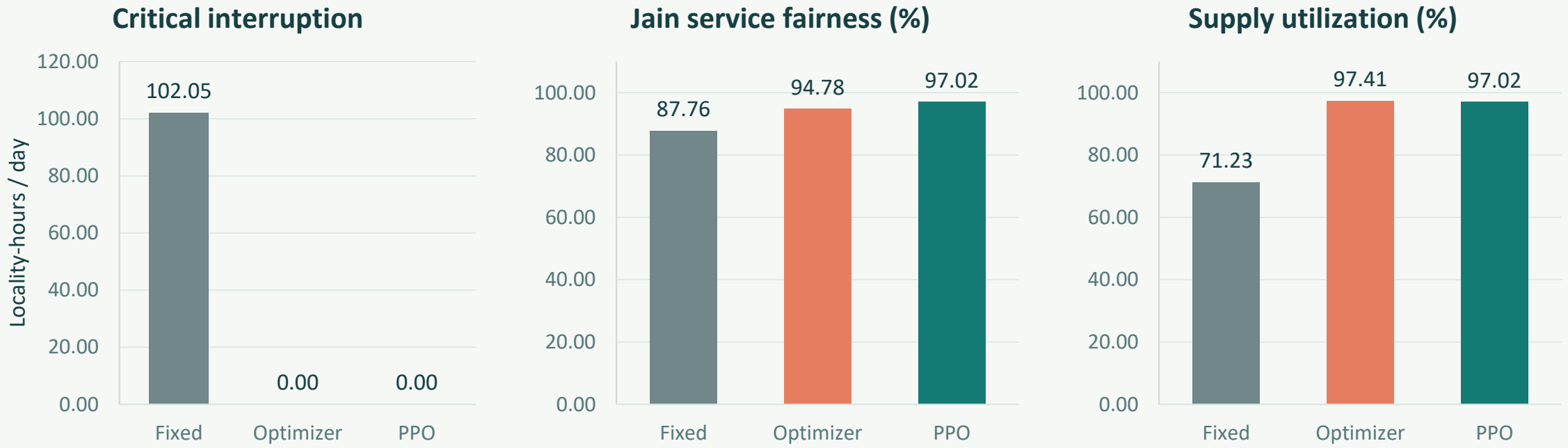
A positive reward difference alone does not satisfy the service guardrails.

Requirement	Exact rule
Adjusted overall intervals	Reward lower bound > 0; fairness and utilization lower bounds ≥ −0.002
Every shortage level	Mean reward difference ≥ 0; fairness and utilization differences ≥ −0.002
Every PPO and optimizer trajectory	Zero critical interruptions, outages and power-balance violations
Validation eligibility	All five selected policies eligible under the frozen selection rule

Use adjusted 97.5% intervals. The 0.002 margin is 0.2 percentage points. The separate +0.01 reward reference is a research choice, not an operational threshold.

Original study on 960 matched conditions

PPO values average the five independently trained policies equally



PPO and the optimizer preserve critical service. Their remaining differences concern allocation.

PPO improves service over Fixed Rotation

Paired effects are PPO minus Fixed Rotation. Intervals have adjusted 97.5% coverage.

Outcome	Effect [lower, upper]
Critical interruption (locality-hours)	-102.050 [-102.081, -102.025]
Jain fairness (percentage points)	+9.260 [+9.142, +9.371]
Utilization (percentage points)	+25.782 [+25.702, +25.859]

Full declared comparison: PASSED

Critical-aware rotation also eliminates these interruptions. The follow-up comparison measures PPO’s additional allocation benefit when critical protection is shared.

A reward and fairness gain with a utilization tradeoff

Paired effects are PPO minus Constrained Optimization. Adjusted 97.5% intervals.

Outcome	Effect [lower, upper]
Daily reward	+0.0499 [+0.0406, +0.0577]
Jain fairness (percentage points)	+2.237 [+2.096, +2.373]
Utilization (percentage points)	-0.399 [-0.464, -0.341]
Switching (percentage points)	+7.218 [+6.531, +8.112]

Full service-guarded improvement criterion: NOT MET

The utilization loss exceeds the allowed 0.2-point margin. Both protect critical service.

Practical scale of the optimizer comparison

Measured energy and public service outcomes give the composite reward context.

Measure	Optimizer	PPO	Difference
Daily reward	23.4468	23.4967	+0.0499
Served energy (MWh/day)	6547.90	6528.02	-19.88
Jain fairness (%)	94.78	97.02	+2.237 pp
Supply utilization (%)	97.41	97.02	-0.399 pp

The +0.0499 reward gain is about 0.208% of the ideal daily ceiling of 24. PPO serves 19.88 MWh less per day while distributing service more evenly.

Duration fairness and total interruption burden

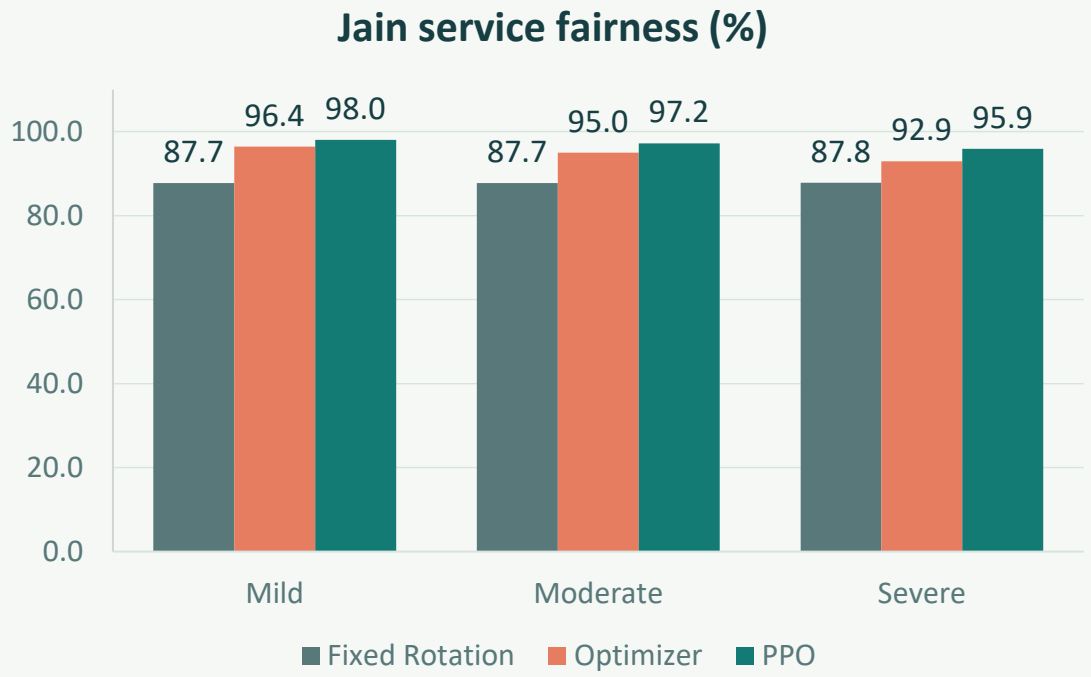
Episode duration measures are calculated before averaging

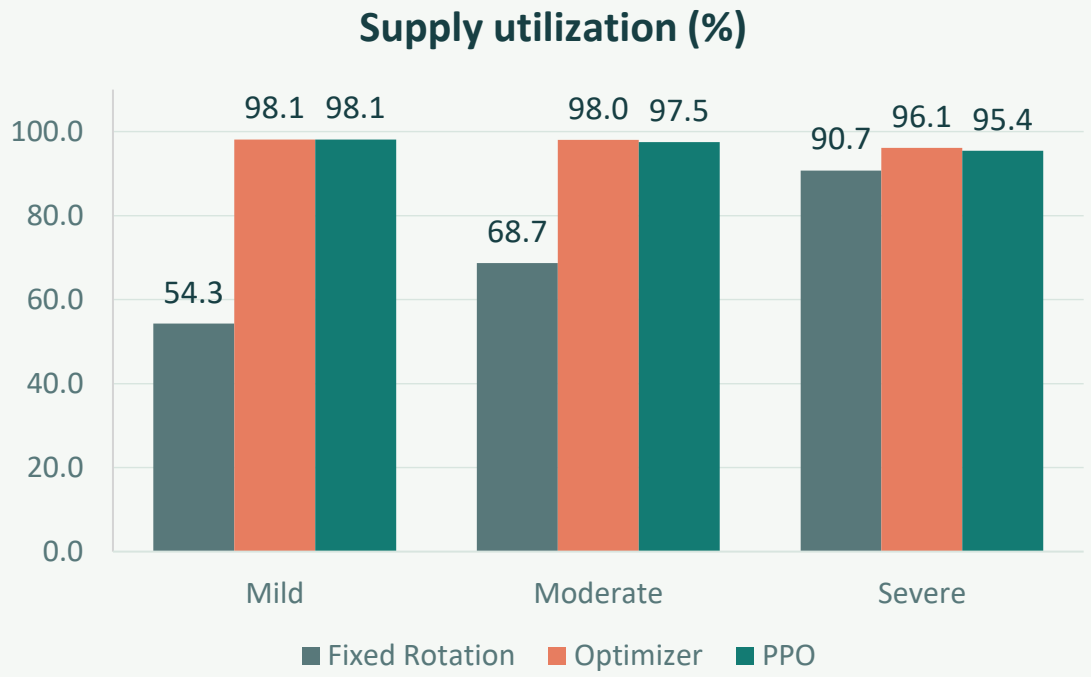
Measure	Fixed Rotation	Optimizer	PPO
Zero-service blackout (locality-hours)	96.00	0.00	0.00
Blackout pairwise gap (hours)	1.71	0.00	0.00
Critical-interruption pairwise gap (hours)	1.90	0.00	0.00
Longest critical interruption (hours)	8.00	0.00	0.00

Critical Only removes locality-wide zero-service blackouts while Residential and Commercial demand can remain unmet. Zero duration does not mean Full Service.

Shortage-specific allocation outcomes

Results keep the Mild, Moderate and Severe conditions visible





Critical interruption is zero for PPO and the optimizer at every tested shortage level.

Final policies against stronger simple baselines

Post-hoc audit: the same 960 final conditions, with no retraining or policy reselection.

Controller	Critical locality-h	Fairness %	Utilization %	Daily reward
Fixed Rotation	102.05	87.76	71.23	13.9688
Critical-aware rotation	0.00	94.95	93.38	23.2257
Fairness Rule	0.00	96.15	97.32	23.4865
Constrained optimizer	0.00	94.78	97.41	23.4468
PPO (five-run mean)	0.00	97.02	97.02	23.4967

Critical-aware rotation reserves critical demand, then fills supply in fixed cyclic priority. Fairness Rule uses the unchanged demonstration-teacher setting.

Learning beyond critical-service protection

PPO minus each baseline. Follow-up intervals are descriptive paired 95% intervals.

Baseline	Served MWh/day	Daily reward	Fairness pp
Critical-aware rotation	+241.63 [+236.89, +245.91]	+0.2710 [+0.2631, +0.2782]	+2.067 [+1.958, +2.172]
Fairness Rule	-16.68 [-20.84, -12.99]	+0.0102 [+0.0024, +0.0172]	+0.864 [+0.747, +0.976]

PPO serves 241.63 MWh/day more than protected rotation. Against its teacher, PPO gains a small amount of reward and fairness but serves 16.68 MWh/day less.

The teacher comparison varies by shortage level and training run. It does not support an unconditional PPO victory.

When critical demand exceeds supply

Stress audit: 320 matched base draws at each of two critical-supply fractions.

PPO outcome	75% of critical demand	25% of critical demand
All-Critical-Only fallback (hours)	24	24
Critical interruption (locality-hours)	168	168
Complete critical outage (locality-hours)	0	0
Unmet critical energy (MWh)	671.51	2014.53
Power-balance violation rate	0	0

All constrained methods share the proportional fallback. Partial service deprivation affects all seven localities for 24 hours; no power limit is exceeded.

Validation progress across all five training runs

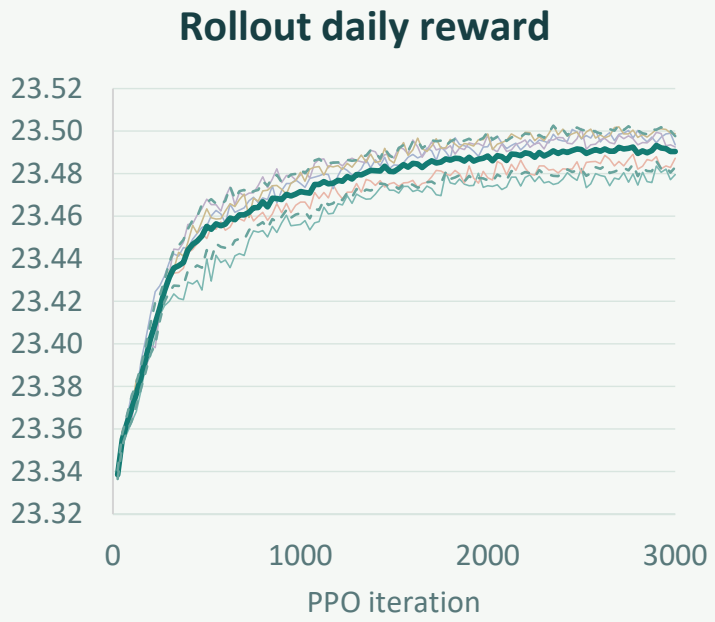
31 checkpoints per run. Checkpoint selection never consults final-test performance.

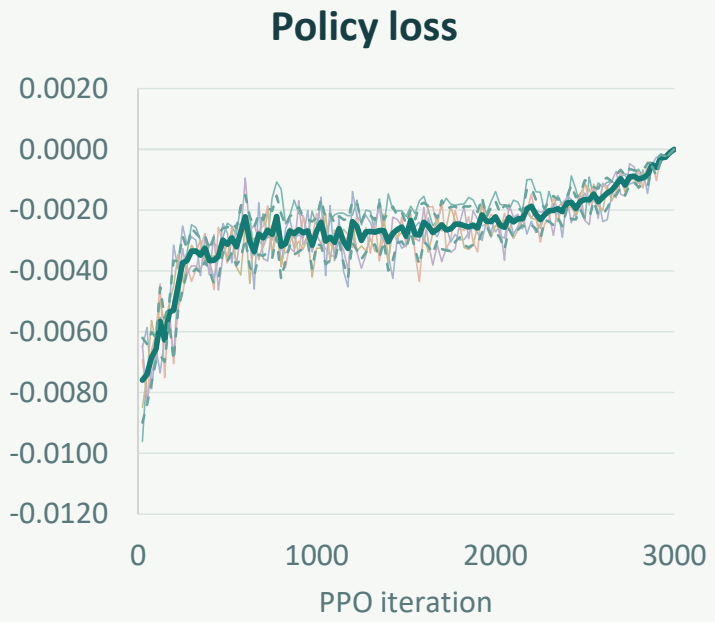


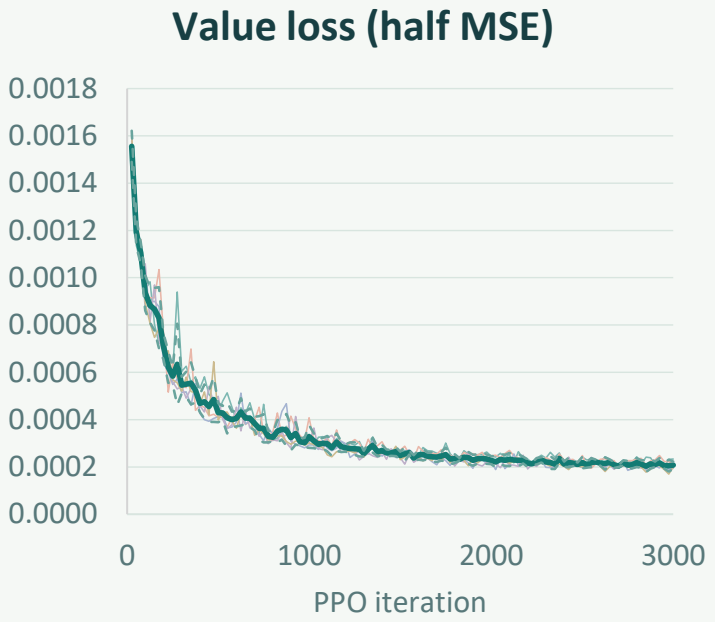
Seeds: 1729 2718 3141 5772 8111

Training reward and loss

120 recorded updates per seed, every 25 iterations. No smoothing.





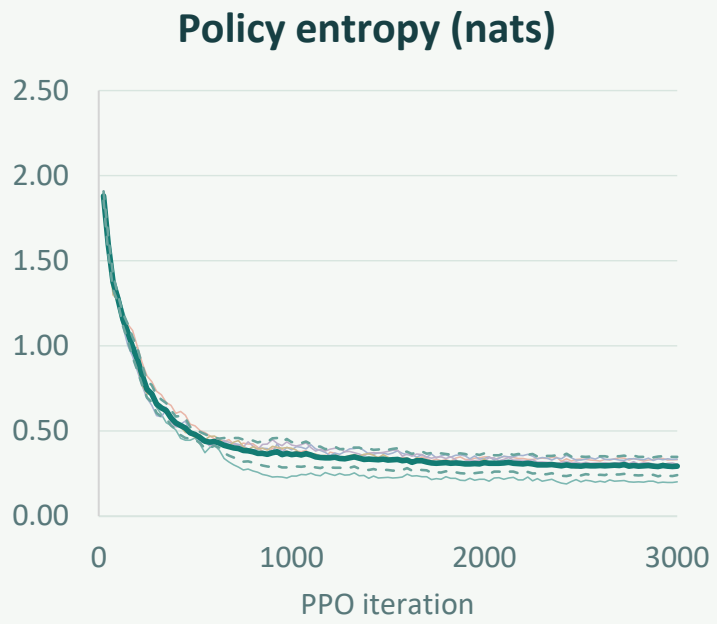


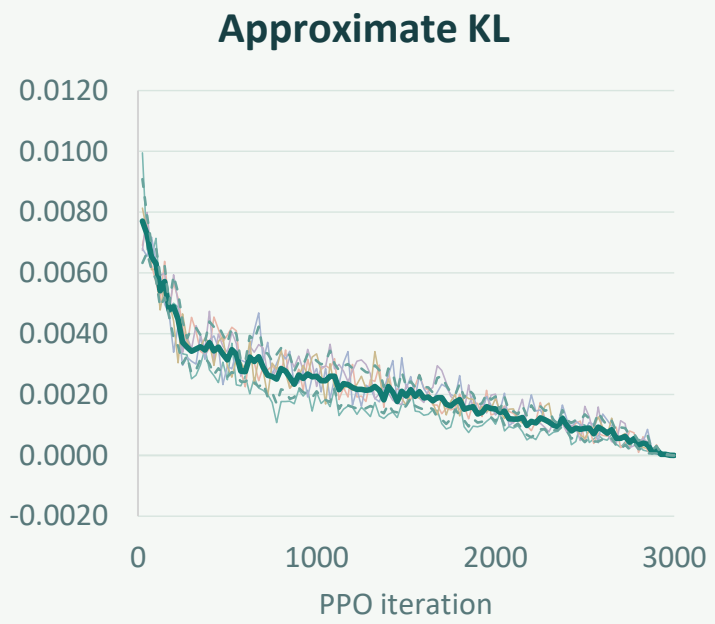
Thin lines: five seeds. Dark line: mean. Dashed limits: mean ± sample SD, not a CI.

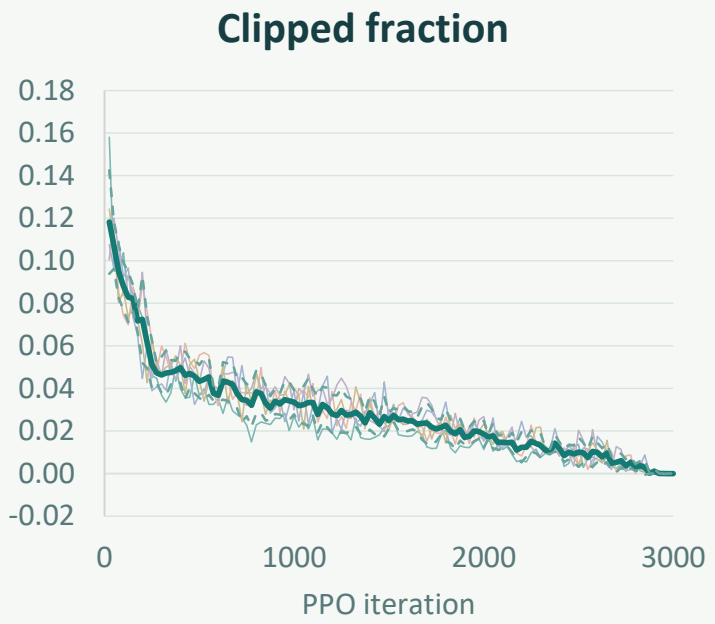
Reward describes the pre-update rollout. Losses average all minibatches of that update’s four epochs.

Policy entropy and update diagnostics

120 recorded updates per seed, every 25 iterations. No smoothing.





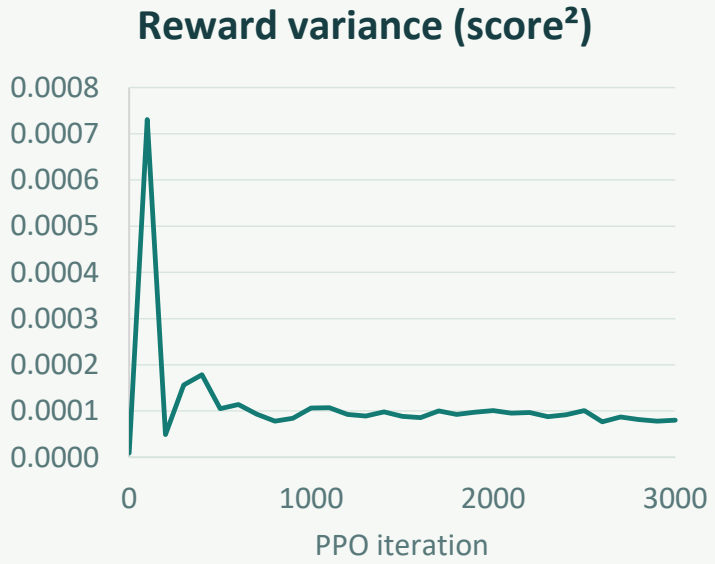


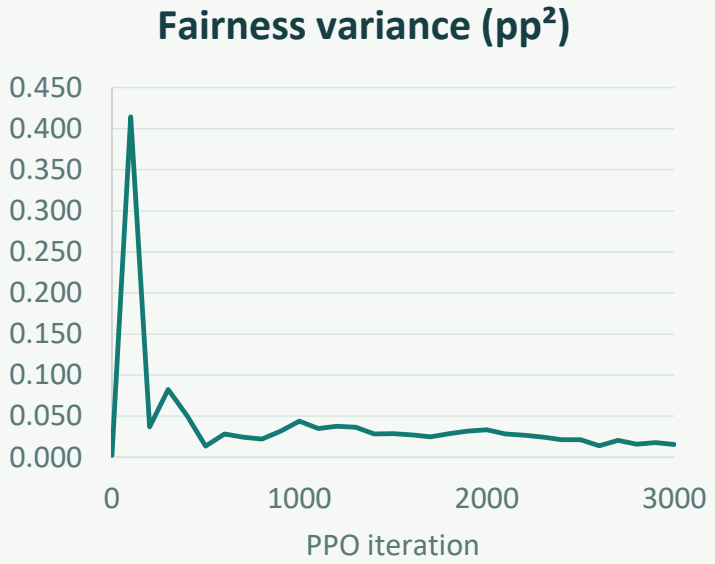
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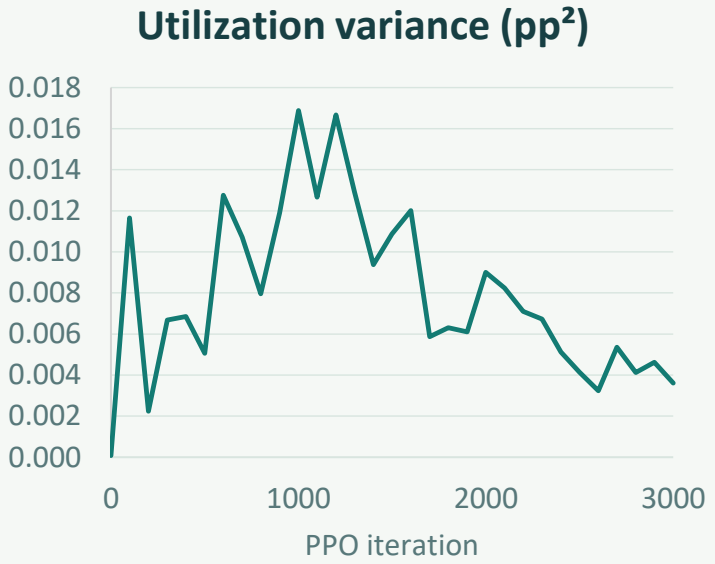
Entropy, KL and clipping average all minibatches of the logged update. They are not 25-update moving averages.

Between-seed validation variance

Sample variance of the five run means on the same validation scenarios







$$\bar{x}_j = \frac{1}{5}\sum_{k=1}^{5} x_{k,j}, \qquad s_j^2 = \frac{1}{4}\sum_{k=1}^{5}(x_{k,j} - \bar{x}_j)^2$$

31 checkpoints include behavior-cloning iteration 0. This is between-run variance, not critic explained variance or final-test uncertainty.

Longer training improves reward but loses utilization

Same-run iteration-600 snapshots are a within-schedule reference

Seed	Selected iteration	Test reward at 600	Selected test reward	Difference
1729	2900	23.4497	23.4835	+0.0338
2718	3000	23.4654	23.4913	+0.0258
3141	2900	23.4699	23.5034	+0.0335
5772	3000	23.4781	23.5014	+0.0233
8111	2500	23.4732	23.5040	+0.0309

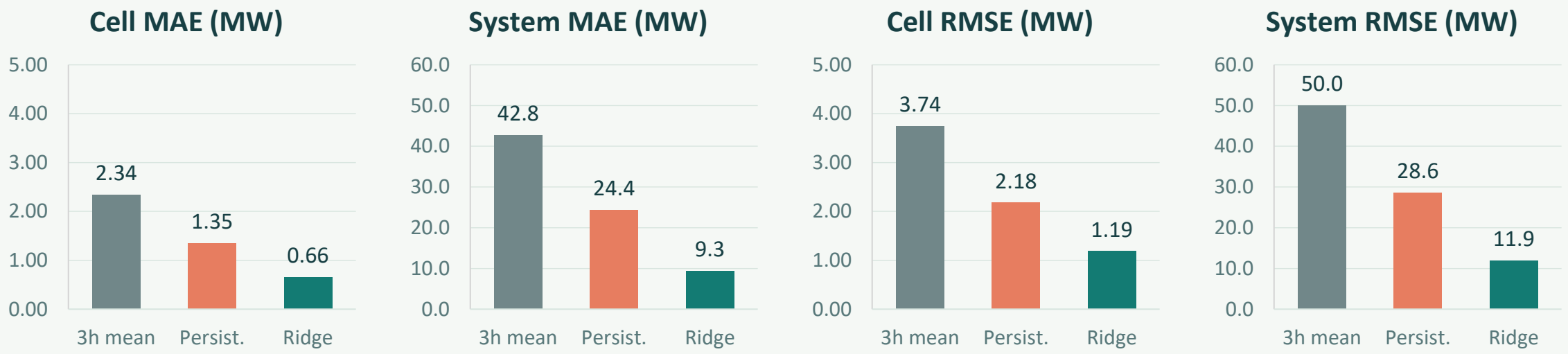
Mean daily reward +0.0295. Utilization −0.2417 percentage points.

The full additional-training criterion fails its utilization and shortage guardrails. The 600 snapshots use the planned 3,000-iteration schedule.

Learned next-hour demand prediction

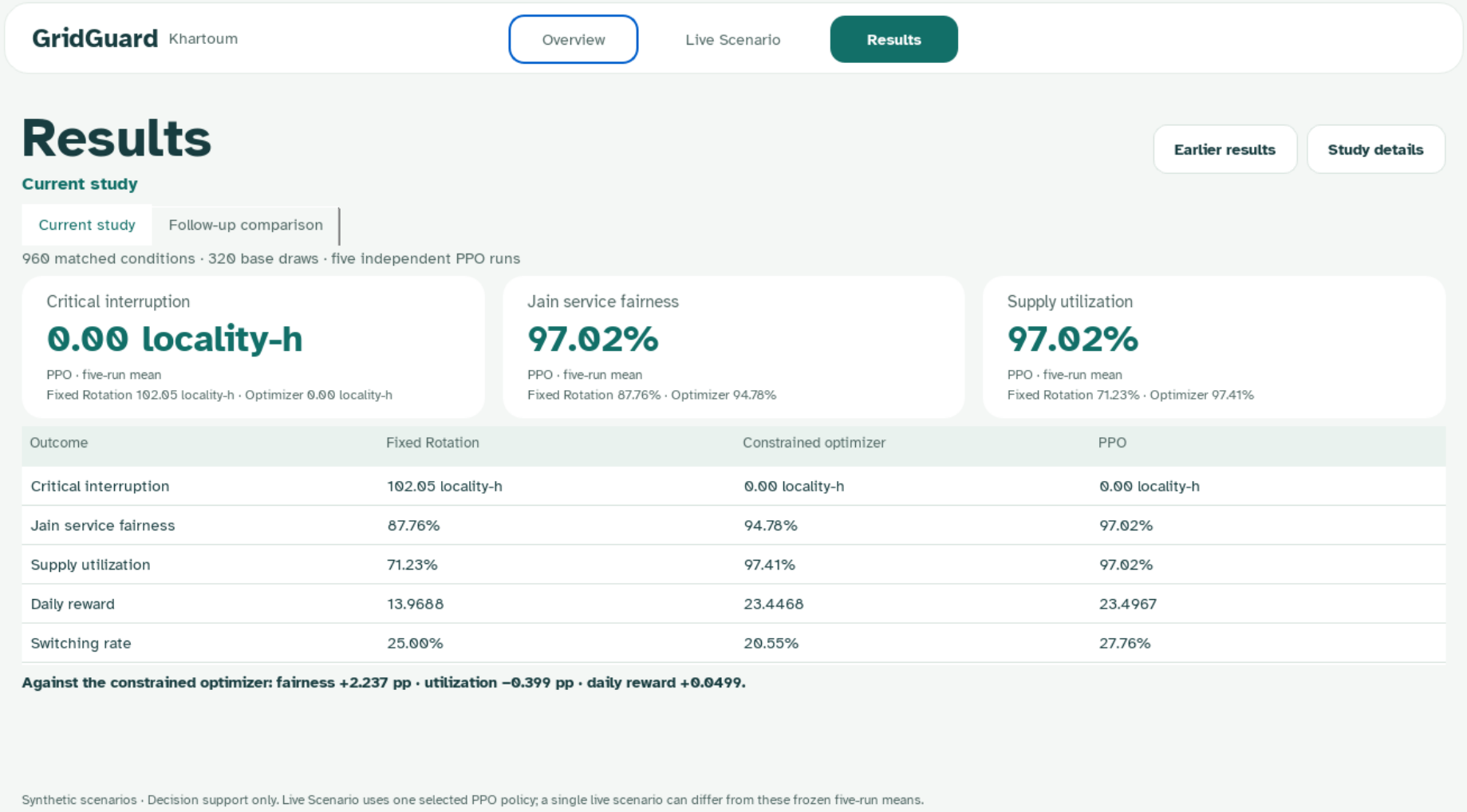
Ridge AR(3): 768 training traces and 320 separate evaluation traces

$$\widehat{d}_t = \max(0, \beta_0 + \beta_1 d_{t-1} + \beta_2 d_{t-2} + \beta_3 d_{t-3})$$



Forecasts use past demand and identical target hours. Prediction does not enter PPO.

Current study results in the desktop app



Conclusions and the next engineering step

Against Fixed Rotation

The constrained PPO system improves critical service, fairness and utilization on fresh matched synthetic scenarios.

Against the optimizer

PPO improves fairness and daily reward but uses less supply and switches more often. The full criterion fails.

- PPO also improves allocation over critical-aware rotation in the follow-up audit.
- The final teacher comparison shows a small reward gain with an energy tradeoff.
- Measured demand and feasible feeder actions are needed before operational evaluation.

References 1 of 2

Alphabetized author-date references

Agarwal, R., Schwarzer, M., Castro, P.S., Courville, A. and Bellemare, M.G. (2021) “Deep reinforcement learning at the edge of the statistical precipice”, *NeurIPS 34*. https://arxiv.org/abs/2108.13264

Chen, H., Zhuang, J., Zhou, G., Wang, Y., Sun, Z. and Levron, Y. (2023) “Emergency load shedding strategy for high renewable energy penetrated power systems based on deep reinforcement learning”, *Energy Reports*, 9, pp. 434–443. https://doi.org/10.1016/j.egyr.2023.03.027

García, J. and Fernández, F. (2015) “A comprehensive survey on safe reinforcement learning”, *JMLR*, 16(42), pp. 1437–1480. https://jmlr.org/papers/v16/garcia15a.html

Huang, S. and Ontañón, S. (2022) “A closer look at invalid action masking in policy gradient algorithms”, *FLAIRS Conference Proceedings*, 35. https://doi.org/10.32473/flairs.v35i.130584

References 2 of 2

Alphabetized author-date references

Jain, R., Chiu, D.-M. and Hawe, W.R. (1984) *A quantitative measure of fairness and discrimination for resource allocation in shared computer systems*. DEC Research Report TR-301.

Schulman, J., Wolski, F., Dhariwal, P., Radford, A. and Klimov, O. (2017) "*Proximal policy optimization algorithms*". https://doi.org/10.48550/arXiv.1707.06347

World Bank (2025) *Accelerating Sustainable and Clean Energy and Digital Access Transformation in Sudan* (P507788). Project Information Document, appraisal stage, PIDIA01286, 1 May. documents.worldbank.org, project P507788.